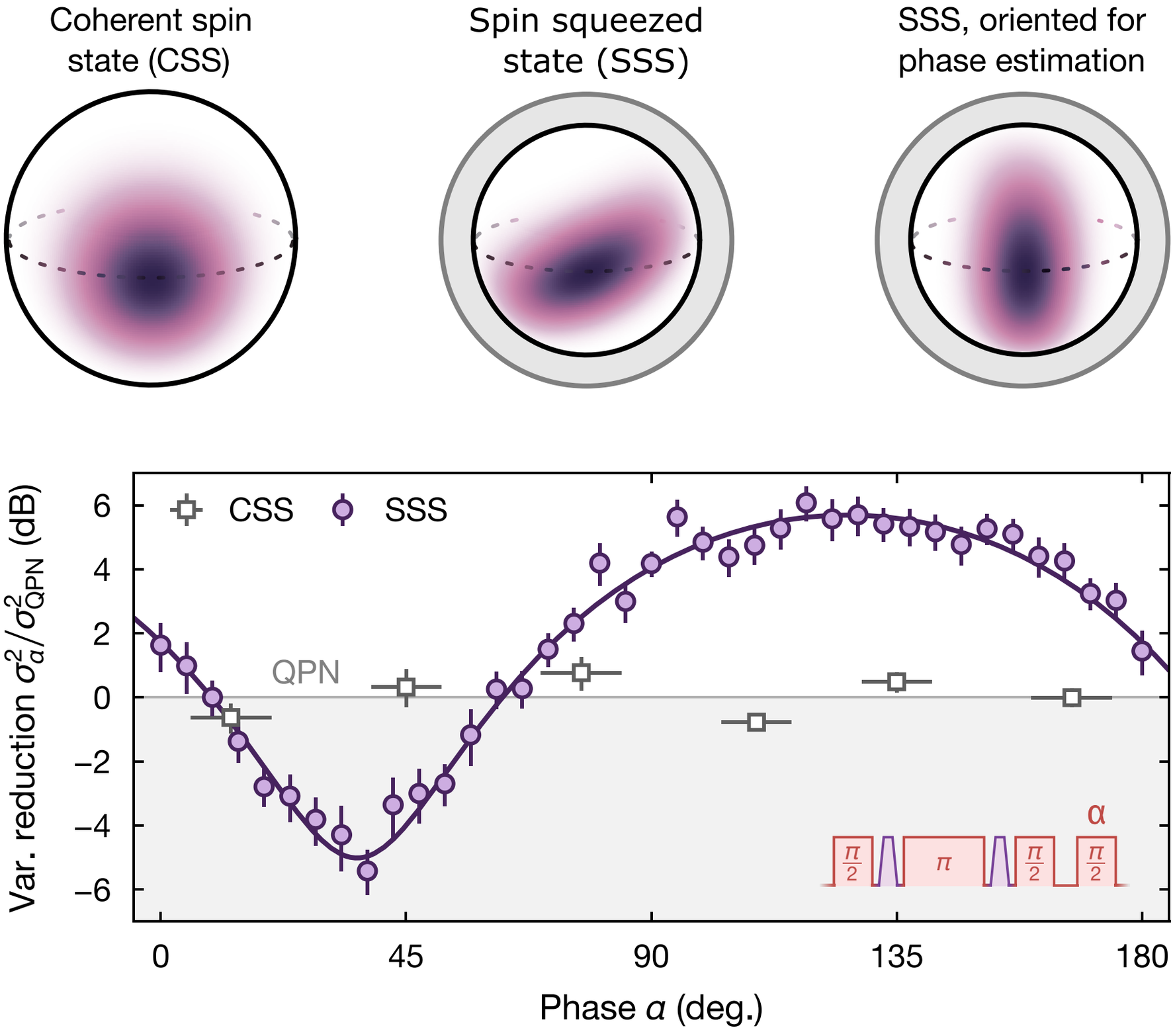

Coherent spin state (CSS)
Spin squeezed state (SSS)
SSS, oriented for phase estimation
CSS
SSS
QPN
6
4
2
0
−2
−4
−6
0
45
90
135
180
Var. reduction $\sigma_\alpha^2/\sigma_{QPN}^2$ (dB)
Phase α (deg.)
α
$\frac{\pi}{2}$
π
$\frac{\pi}{2}$
$\frac{\pi}{2}$